

# Navigating the Privacy-Robustness-Performance-Trilemma in Federated Learning

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# Agenda

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## Desirable Properties of Federated Learning

| Property           | <sup>CCC</sup><br>Provides                                | Cost                                |
|--------------------|-----------------------------------------------------------|-------------------------------------|
| <b>Privacy</b>     | Limited information leakage, protection of client data    | Accuracy loss<br>slower convergence |
| <b>Robustness</b>  | Decreased vulnerability to Byzantine or poisoning attacks | Gradient distortion → Accuracy loss |
| <b>Performance</b> | Increased model utility and efficiency                    | drawbacks in privacy or robustness  |